

Efficiency Improving Lobbying and Markup

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Abstract

This paper examines how lobbying affects economic efficiency through a dynamic model of endogenous policy influence. Firms with heterogeneous productivity engage in Cournot competition and may lobby for output subsidies, with larger, more productive firms lobbying more intensively. While conventional views emphasize lobbying's distortionary effects, the model shows that it can enhance aggregate productivity by reallocating resources toward underproducing high-productivity firms. The productivity gains are most pronounced in competitive industries, where lobbying disproportionately benefits efficient producers. However, lobbying also creates inter-industry distortions by favoring concentrated sectors regardless of their efficiency. The model further explains rising markup skewness, as subsidies expand output from productive firms while compressing rivals' margins. Preliminary calibration suggests that lobbying increases GDP by 1.4%, indicating net welfare gains from correcting misallocation.

1 Introduction

To what extent does lobbying shape economic policy and market outcomes? In the 110th US Congress, only 4.4% of energy-related bills were enacted without lobbying support (Kang 2016), suggesting that lobbying can be viewed as a decentralized mechanism for resource allocation, alongside voting and campaign financing. Conventional wisdom holds that lobbying distorts free markets by granting artificial advantages to certain interest groups, leading to economic inefficiencies. However, in imperfect economies—where initial policies may be suboptimal or firms face misallocation—the welfare implications of lobbying become ambiguous. If lobbying efforts correlate with firm productivity, they could potentially mitigate existing distortions by reallocating resources toward more efficient producers.

This paper develops a model in which firms with heterogeneous productivity engage in Cournot competition within industries producing homogeneous goods. In this setting, highly productive firms underproduce, while less productive firms overproduce rel-

ative to the social optimum. Firms may endogenously lobby for output subsidies, with larger, more productive firms lobbying more intensively. As a result, lobbying redistributes market share toward these firms, raising aggregate industry productivity. The productivity-enhancing effect of lobbying grows stronger with the level of competition within an industry, implying that more competitive industries benefit disproportionately from lobbying activity.

However, lobbying also generates inter-industry distortions. Industries with fewer firms produce more on individual firm level, leading to higher markups and stronger incentives to lobby. Resources may flow toward naturally concentrated industries not due to their productivity. This creates a countervailing distortion, where lobbying-induced overproduction in industries with less firms may offset intra-industry productivity gains.

The model further explains the rising skewness in markups observed in recent decades. Subsidies expand output from high-productivity firms, lowering equilibrium prices and squeezing markups for less efficient competitors. At the same time, subsidized firms see their own markups rise, widening the dispersion between industry leaders and laggards.

A preliminary calibration suggests that lobbying increases aggregate productivity, raising GDP by 1.4%. While the full quantitative exercise remains incomplete, these results indicate that lobbying may generate net welfare gains by correcting preexisting misallocation. The paper concludes with a discussion of how lobbying parameters influence efficiency and the potential design of optimal lobbying mechanisms.

Literature Review

There are three main strands of literature relevant to this paper. The first strand is the auction-based literature on lobbying. Bernheim and Whinston (1986) model lobbying as a first-price menu auction, while Baye et al. (2008) consider it as an all-pay auction. Extending this framework, Fang (2002), Chowdhury and Topolyan (2016), and Matsueda (2020) analyze various auction-based lobbying models under different institutional rules. This literature generally emphasizes efficiency in lobbying outcomes, as scarce legislative opportunities tend to be allocated to those entities that value them most highly. The present paper does not explicitly model lobbying as an auction; instead, legislative scarcity emerges endogenously through the lobbying cost structure. Nevertheless, efficiency outcomes resembling auction results do arise within this model.

The second strand of literature treats lobbying primarily as a distortionary force within otherwise efficient market structures. For example, Huneus and Kim (2021) develop a general equilibrium framework in which lobbying affects firm-specific taxation policies. In their model, all firms are identical, and a uniform tax is optimal. Unlike their approach, my paper incorporates firm heterogeneity in productivity, implying that uniform taxation is not optimal and that the direction and magnitude of distortionary

effects become critically important.

Empirical evidence further underscores the relevance of lobbying-induced distortions. De Loecker et al. (2020) document rising markups and increased skewness over recent decades. Kim (2018) provides evidence linking increased lobbying expenditure to these changes in markup distributions. Richter et al. (2009) and Kerr et al. (2014) offer comprehensive linkages between firm-level lobbying activities and corporate financial characteristics, emphasizing the strong persistence and firm-size dependency of lobbying behaviors. Specifically, Kerr et al. (2014) highlight that lobbying persistence is driven largely by substantial initial fixed costs incurred by firms establishing their political presence. This feature aligns closely with my model, which incorporates fixed lobbying costs and lobbying induced markup.

This paper builds upon Edmond et al. (2023), who used nested competition structures within a CES framework. Incorporating their approach, my paper explores how productivity heterogeneity, industry structure, and lobbying interact to influence firm-level outcomes and aggregate efficiency. Consistent with Kang (2016) and Bombardini and Trebbi (2011, 2012), the structural econometric literature supports modeling lobbying decisions explicitly, capturing how lobbying investments translate into policy outcomes.

Furthermore, Richter et al. (2009) specifically investigate lobbying's tax implications, demonstrating that higher lobbying expenditures significantly reduce firms' effective tax rates. According to their estimates, a 1% increase in lobbying expenditures corresponds to a 0.5 to 1.6 percentage point reduction in effective tax rates. I model lobbying as an effective tax cut.

2 Model

The economy features a representative household that maximizes lifetime utility from consumption of a final good while supplying labor services and accumulating physical capital. Production is organized through three vertically related sectors: (1) a perfectly competitive final goods sector that aggregates intermediate inputs using CES technology, (2) multiple differentiated intermediate goods industries, and (3) oligopolistic firms within each industry that produce homogeneous goods under Cournot competition. Firms engage in simultaneous strategic decisions regarding both production quantities and political lobbying expenditures, where lobbying generates firm-specific output subsidies subject to convex costs and persists for only the current period. Productivity evolves exogenously through independent and identically distributed Pareto shocks received by each firm every period. The capital stock accumulates through optimal investment decisions that converge to a unique steady state level. The equilibrium concept is a competitive and Nash equilibrium where all firms optimize production and lobbying decisions taking competitors' strategies as given, while factor markets clear and the representative household

maximizes intertemporal utility.

Household Optimization Problem

The representative consumer solves the infinite-horizon optimization problem:

$$\max_{\{C_t, L_t, K_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \left(\log C_t - \psi_l \frac{L_t^{1+\phi_L}}{1+\phi_L} \right) \quad (1)$$

subject to the sequential budget constraint:

$$C_t + K_{t+1} \leq w_t L_t + (1 + r_t - \delta) K_t + \sum_{i=1}^N \sum_{j=1}^{M_i} \pi_{ij,t} + \Pi_t + T_t \quad (2)$$

where C_t denotes consumption of the final good, L_t represents labor supply, K_t is the physical capital stock, w_t equals the real wage rate, r_t denotes the rental rate of capital, $\pi_{ij,t}$ captures profits from firm j in industry i , Π_t represents final goods producer profits, and T_t stands for lump-sum government transfers. The household's optimal labor supply decision satisfies the first-order condition:

$$\psi_l L_t^{\phi_L} = w_t \quad (3)$$

while the intertemporal Euler equation governing capital accumulation takes the form:

$$1 = \beta \frac{C_t}{C_{t+1}} (r_{t+1} + 1 - \delta) \quad (4)$$

Final Goods Production

The perfectly competitive final goods sector combines differentiated intermediate goods from N distinct industries through a constant elasticity of substitution (CES) production technology:

$$Y_t = \left(\sum_{i=1}^N \theta_i Q_{it}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad (5)$$

where Y_t represents final output (with price normalized to unity), Q_{it} denotes the composite good from industry i , $\sigma > 1$ equals the elasticity of substitution across industries, and θ_i are industry-specific weights satisfying $\sum_{i=1}^N \theta_i = 1$. The final goods producer solves the static profit maximization problem:

$$\max_{\{Q_{it}\}} \Pi_t = Y_t - \sum_{i=1}^N p_{it} Q_{it} \quad (6)$$

yielding the following first-order condition for each intermediate good:

$$p_{it} = \theta_i \left(\frac{Y_t}{Q_{it}} \right)^{1/\sigma} P_t \quad (7)$$

The aggregate price index consistent with the CES aggregator is given by:

$$P_t = \left(\sum_{i=1}^N \theta_i^\sigma p_{it}^{1-\sigma} \right)^{\frac{1}{1-\sigma}} \quad (8)$$

Industry-Level Production and Firm Behavior

Each industry i consists of M_i firms engaged in Cournot quantity competition. Firms simultaneously choose production levels q_{ijt} and lobbying expenditures l_{ijt} , taking as given the actions of other firms in the economy. Production technologies exhibit constant returns to scale:

$$q_{ijt} \leq a_{ijt} n_{ijt}^\alpha k_{ijt}^{1-\alpha} \quad (9)$$

where n_{ijt} denotes labor input, k_{ijt} represents capital input, and a_{ijt} captures firm-specific productivity drawn each period from a Pareto distribution with shape parameter ξ and scale parameter $a_{\min} = 1$.

Lobbying activities generate output subsidies according to the technology:

$$l_{ijt}^{-1}(s) = \begin{cases} 0 & \text{if } l_{ijt} = 0 \\ c_s + \psi_s s^{\phi_s} & \text{if } l_{ijt} > 0 \end{cases} \quad (10)$$

where $\phi_s > 1$ ensures strictly convex costs of lobbying and $c_s > 0$ represents a fixed cost incurred whenever $l_{ijt} > 0$. The convexity reflects increasing marginal costs of political influence, while the fixed cost captures necessary institutional investments for effective lobbying.

Firm Optimization Problem

Each firm j in industry i maximizes current-period profits:

$$\pi_{ijt} = [p_{it}(Q_{it}) + s(l_{ijt})]q_{ijt} - w_t n_{ijt} - r_t k_{ijt} - l_{ijt} \quad (11)$$

subject to the production technology constraint (9). The optimization problem decomposes naturally into two stages. First, conditional on output level q_{ijt} , the firm minimizes costs by solving:

$$\min_{n_{ijt}, k_{ijt}} w_t n_{ijt} + r_t k_{ijt} \quad (12)$$

$$\text{s.t. } a_{ijt} n_{ijt}^\alpha k_{ijt}^{1-\alpha} \geq q_{ijt} \quad (13)$$

yielding conditional factor demands:

$$n_{ijt} = \left(\frac{q_{ijt}}{a_{ijt}} \right) \left(\frac{r_t \alpha}{w_t (1 - \alpha)} \right)^{1-\alpha} \quad (14)$$

$$k_{ijt} = \left(\frac{q_{ijt}}{a_{ijt}} \right) \left(\frac{r_t \alpha}{w_t (1 - \alpha)} \right)^{-\alpha} \quad (15)$$

Substituting these into the profit function yields the reduced-form optimization problem in (q_{ijt}, l_{ijt}) space:

$$\max_{q_{ijt}, l_{ijt}} \left[p_{it}(Q_{it}) + s(l_{ijt}) - \frac{MC_t}{a_{ijt}} \right] q_{ijt} - l_{ijt} \quad (16)$$

where $MC_t \equiv w_t^\alpha r_t^{1-\alpha} [\alpha^{-\alpha} (1 - \alpha)^{-(1-\alpha)}]$ represents the common marginal cost component across all firms.

General Equilibrium Conditions

The model's recursive competitive equilibrium consists of:

- Price functions $\{w_t, r_t, \{p_{it}\}_{i=1}^N\}$
- Allocation rules $\{C_t, L_t, K_{t+1}, \{Q_{it}, \{q_{ijt}, n_{ijt}, k_{ijt}, l_{ijt}\}_{j=1}^{M_i}\}_{i=1}^N\}$

such that:

1. Final goods market clearing:

$$C_t = Y_t - \sum_{i=1}^N \sum_{j=1}^{M_i} c_g \mathbb{1}(l_{ijt} > 0) \quad (17)$$

2. Labor market clearing:

$$L_t = \sum_{i=1}^N \sum_{j=1}^{M_i} n_{ijt} \quad (18)$$

3. Capital market clearing:

$$K_t = \sum_{i=1}^N \sum_{j=1}^{M_i} k_{ijt} \quad (19)$$

4. Government budget balance:

$$T_t = \sum_{i=1}^N \sum_{j=1}^{M_i} (l_{ijt} - s(l_{ijt}) q_{ijt} - c_g \mathbb{1}(l_{ijt} > 0)) \quad (20)$$

Industry Productivity and Market Power

Industry-level total factor productivity aggregates firm-specific efficiencies through:

$$a_{it} = \frac{Q_{it}}{\sum_{j=1}^{M_i} (q_{ijt}/a_{ijt})} \quad (21)$$

This measure increases when production becomes concentrated among more efficient firms. Firm markups incorporate both market power and lobbying effects:

$$\mu_{ijt} = \frac{(p_{it} + s(l_{ijt}))a_{ijt}}{MC_t} \quad (22)$$

The steady-state equilibrium combines: (1) optimal capital accumulation determined by (4), (2) Nash equilibrium in firm strategies solving (16), and (3) market clearing in all goods and factor markets.

3 Analysis

This study employs a comparative welfare analysis to examine the economic implications of lobbying activities. The investigation proceeds systematically through three distinct analytical stages. First, I characterize the social planner's optimal allocation, establishing the first-best benchmark for economic efficiency. Second, I conduct a comprehensive comparison between this first-best solution and the decentralized equilibrium with lobbying, identifying both the efficiency-restoring and distortionary effects of lobbying mechanisms. Finally, I extend the analysis to consider the constrained optimization problem faced by a planner limited to designing only the parameters of the lobbying mechanism, thereby providing directions to second-best optimal policy framework.

3.1 First-Best Planner's Solution

The welfare analysis in this framework distinguishes between two fundamental approaches to resource allocation: the first-best and second-best solutions. The first-best allocation represents an idealized benchmark where the social planner possesses complete control over all economic resources and operates without any informational constraints. In this scenario, the planner can directly allocate factors of production and determine output levels across all firms and industries to maximize social welfare.

The second-best allocation introduces constraints on the planner's ability to directly control resources. Here, the second best scenario refer to that the planner can only control the parameters related to lobbying to induce firm behaviors. This constrained scenario reflects either a government that can only implement flat-rate taxes without firms' active participation, or one that faces costs in designing an incentive-compatible taxation system that would induce firms to truthfully reveal their productivity.

Under the transversality condition, the planner's problem reduces to maximizing static utility given steady-state capital. Formally, the planner solves:

$$\max_{\{n_{ij}, k_{ij}\}} U(C, L) \quad (23)$$

subject to the following constraints:

$$Y = C = \left(\sum_i \left(\theta_i Q_i^{\frac{\sigma-1}{\sigma}} \right) \right)^{\frac{\sigma}{\sigma-1}} \quad (24)$$

$$Q_i = \sum_j q_{ij} \quad (25)$$

$$q_{ij} = a_{ij} n_{ij}^\alpha k_{ij}^{1-\alpha} \quad (26)$$

$$L = \sum_{ij} n_{ij} \quad (27)$$

$$K^* = \sum_{ij} k_{ij} \quad (28)$$

where individual firm productivities $\{a_{ij}\}$ and consequently the number of firms in each industry are predetermined. The steady-state capital stock K^* satisfies the modified Euler condition:

$$\frac{\partial Y}{\partial k_{ij}} = \frac{1}{\beta} + \delta - 1 \quad \forall i, j \quad (29)$$

The planner's solution exhibits a nested structure that can be decomposed into three sequential steps. First, the planner determines optimal firm-level outputs $\{q_{ij}^*\}$. Second, these outputs are aggregated to industry outputs $\{Q_i^*\}$. Finally, the planner allocates final output Y according to the labor-leisure condition:

$$\frac{\partial Y / \partial n_{ij}}{C} = \psi_l L^{\phi_l} \quad \forall i, j \quad (30)$$

On the final producer level, the optimal ratio of industry outputs satisfies:

$$\frac{Q_i^*}{Q_{i'}^*} = \left(\frac{\theta_i}{\theta_{i'}} \right)^\sigma \left(\frac{a_{i'}}{a_i} \right)^\sigma \quad (31)$$

where a_i represents the industry level productivity for i . More valued industry goods and less costly industry goods gets produced more.

At the industry level, given fixed Q_i the planner's minimize production cost across firms:

$$\min_{\{q_{ij} \geq 0\}} \sum_{j=1}^{M_i} \frac{q_{ij}}{a_{ij}} \quad (32)$$

Under the assumption of homogeneous goods and constant returns to scale (CRS)

production technologies, the optimal allocation exhibits complete specialization: only the firm with the highest productivity $a_i^{max} = \max_j \{a_{ij}\}$ should produce, with all other firms setting $q_{ij} = 0$ for $a_{ij} < a_i^{max}$. This result follows directly from the linearity of the objective function in (32).

When we introduce either increasing marginal cost, either through diminishing returns to scale (DRS) or firm owned capital, the optimal allocation changes qualitatively. In these cases, the planner's solution involves production by multiple firms, with more productive firms receiving larger output allocations. The exact distribution of production across firms depends on the specific form of the production technology and any additional constraints.

At the factor input level, firms face the same cost minimization problem as the decentralized firm, but with shadow prices reflecting the planner's marginal valuations:

$$w = \frac{\partial F(n_{ij}, k_{ij})}{\partial n_{ij}} \quad \text{and} \quad r = \frac{\partial F(n_{ij}, k_{ij})}{\partial k_{ij}} \quad (33)$$

3.2 Industry Productivity Increase

The social planner's first-best solution requires complete specialization, with only the most productive firm in each industry producing output. However, under Cournot competition with endogenous lobbying, this efficient allocation is rarely achieved with finite amount of firms.

The first-order condition for quantity choice reveals the fundamental trade-offs:

$$p'_i(Q)q_{ij}^* + p_i(Q) + s(l_{ij}^*) = \frac{MC}{a_{ij}} \quad (34)$$

This condition can be rearranged to express the equilibrium quantity:

$$q_{ij}^* = \frac{\frac{MC}{a_{ij}} - p_i(Q) - s(l_{ij}^*)}{p'_i(Q)} \quad (35)$$

Within a given industry, all firms face identical demand conditions, implying common values for both the price p_i and its derivative p'_i . The differences across firms decision emerge from two sources: their i.i.d. productivity levels a_{ij} and their chosen lobbying expenditures l_{ij}^* . The negative slope of the demand function ($p'_i < 0$), which arises from the final producer FOC, implies firms with higher productivity (a_{ij}) and lobbying (l_{ij}^*) produce more output.

The Cournot equilibrium with lobbying exhibits have the most productive firms systematically under-produce relative to the social optimum, while less productive firms over-produce. This misallocation creates deadweight loss and reduces industry level productivity.

$$\mu_{ij} = \frac{(s_{ij} + p_i)a_{ij}}{MC} \quad (36)$$

The model demonstrates that lobbying can partially offset the distortions from Cournot competition when higher productivity firms engage more intensively in lobbying activities. This relationship emerges through two complementary channels in equilibrium. First, the quantity choice equation shows that more productive firms naturally produce more output ($\partial q_{ij}^*/\partial a_{ij} > 0$). Second, the first-order condition for optimal lobbying decisions establishes that firms with greater output have stronger incentives to lobby:

$$1 = q_{ij}^* s'(l_{ij}^*) = \frac{q_{ij}^* [s(l_{ij}^*)]^{1-\phi_s}}{\psi_s \phi_s} \quad (37)$$

The fixed cost threshold ($l_{ij}^* > 0$ when $q_{ij}^* \geq c_s$) ensures that only sufficiently large firms engage in lobbying. When $\phi_s > 1$, this creates a self-reinforcing cycle where higher output induces more lobbying, which through the subsidy effect leads to further output expansion. This prediction aligns with empirical evidence showing that larger, more productive firms tend to lobby more intensively.

One's markup has to increase if a firm chooses to lobby than not in the equilibrium. Marginal cost is constant. When industry output remains fixed, price p_i stays constant, allowing the subsidy s_{ij} to directly increase markups. With endogenous output adjustments, firms will only lobby if the combined effect $s(l_{ij}) + p_i$ covers at least the fixed cost of lobbying. This can only be true if $s(l_{ij} > 0) + p_i(Q_i|l_{ij} > 0) > p_i(Q_i|l_{ij} = 0)$. In other words, since lobbying incurs a fix cost, when marginal cost stays constant, the firm will only choose to lobby if the resulting price increases rather than decreases so profit increase.

As the number of industries grows large, the model predicts that increased lobbying by any firm reduces competitors' markups through industry output expansion. Total industry output Q_i has to rises after a firm endogenously switch to lobbying, due to the FOC and SOC of firm's decision. This output expansion depresses the equilibrium price $p_i(Q)$, while competitors' marginal costs and productivities remain unchanged. Consequently, their markups decrease as shown by:

$$\frac{\partial \mu_{-ij}}{\partial l_{ij}} = \frac{a_{-ij}}{MC} \frac{\partial p_i}{\partial Q_i} \frac{\partial Q_i}{\partial l_{ij}} < 0 \quad (38)$$

These results suggest that lobbying activity polarize markups within industries. Successful lobbying firms achieve higher markups through direct subsidy effects, while non-lobbying competitors experience margin compression from the resulting price depression. This mechanism provides a theoretical foundation for understanding increased markup dispersion observed in many industries, and provides a narrative on how differential lobbying activity can contribute to growing inequality in markup distribution.

3.3 Inter-Industry Distortion

Cournot competition is a special case of the CES framework when the elasticity of substitution, σ , approaches infinity. Consequently, the productivity-driven reallocation effects observed within industries under CES also apply, though less intensely, at the inter-industry level. However, industries differ not only in productivity levels but also in the number of firms they contain. It turns out that distortions arising from variations in firm numbers often overshadow those stemming from productivity heterogeneity.

Lobbying activity is directly associated with output quantity. Under the CES framework, productivity influences quantity, but the number of firms affects quantity even more significantly. While lobbying generally increases both markups and industrial concentration, the magnitude and characteristics of these effects systematically depend on the number of firms in each industry.

In Cournot markets, an increase in the number of firms raises total output but reduces output per firm. This reduction in individual firm output can be substantial, particularly when the initial number of firms is low, potentially exceeding output variations due to productivity differences across industries. Because lobbying intensity depends on output levels, industry-level variations in firm counts—which are exogenously assigned—exacerbate distortions between industries with fewer firms and those with more firms.

In monopolistic industries, with only a single operating firm, lobbying-induced subsidies do not enhance industry-level efficiency, as there are no competing firms for resource reallocation. Nevertheless, monopolists are likely to lobby when their production quantity surpasses the fixed lobbying cost threshold, leading to increased production. However, this increase reflects only an expansion in output rather than genuine productivity improvements at the industry level.

Conversely, industries with multiple firms have greater potential to achieve productivity gains through lobbying. The presence of multiple producers creates opportunities for reallocating market shares toward more productive firms. However, this potential is offset by the fact that individual firms in more competitive industries produce smaller quantities, reducing their incentives to invest in costly lobbying activities.

This trade-off results in a non-monotonic relationship between the number of firms in an industry and the effectiveness of lobbying. The model suggests that moderately concentrated industries—those with sufficient firms to facilitate meaningful reallocation without excessively diluting individual incentives—may experience the most significant productivity gains from lobbying.

The differing impacts across industries generate two distinct patterns of resource misallocation. First, within industries, lobbying can enhance allocative efficiency by shifting production toward more productive firms. Second, between industries, lobbying creates

distortions by altering relative prices in ways that uncorrelated to productivity differences.

The source of this distortion lies in the differential response of quantity decisions at different aggregation levels. Within industries, production quantities closely align with firm productivity. Across industries, however, quantities correlate more strongly with the inverse of the number of firms. Because firm numbers are mostly perpendicular to productivity levels, this discrepancy creates a divergence between private and social returns to lobbying.

3.4 Input Output Distortion

In general, increases in aggregate productivity lead to higher input prices. However, in certain cases—such as a monopoly industry engaging in lobbying—production can rise disproportionately relative to productivity improvements. Due to an artificial output subsidy, imagine it in a representative firm scenario, firms tend to overproduce, inflating input prices above their efficient levels. Within the model, this distortion results in excess employment and increased savings. Nevertheless, this outcome need not be a feature of lobbying. An output subsidy is chosen as a tool that gives a firm preferential treatment against others, but to provide a net subsidy to all firms. For instance, if output subsidies are financed by output taxes imposed on all firms, such input/output distortions may be reduced.

3.5 Exogenous Exits

Endogenous firm exits are implicitly incorporated in the model. Firms always retain the option to produce zero output. This option becomes particularly relevant when goods within an industry are highly substitutable, there are numerous firms, and productivity draws exhibit substantial variance. Under these conditions, lobbying may increase aggregate output (Q_i) sufficiently to lower the industry price (p_i), causing marginal revenue to fall below the marginal cost for the least productive firms. Such endogenous exits generally enhance efficiency rather than reduce it, as market shares shift toward more productive firms. Consequently, aggregate industry output (Q_i) is at least weakly increased; otherwise, the exiting firms would have remained in operation.

$$q_{ij} = 0 \quad \text{if} \quad p_i(Q_i) + s(l_{ij}) < \frac{MC}{a_{ij}} \quad (39)$$

This exit mechanism has efficiency implications that differ from standard models of firm turnover. Because exit is driven by the combination of lobbying-induced output expansion and productivity differences, it reallocate market share toward more productive firms. The resulting industry equilibrium features weakly higher total output Q_i and improved average productivity. This contrasts with models where exit reduces industry

output.

An import distinction is lobbying driven exit versus where the firm does not exist to begin with. Even in this model, ex ante to productivity realization, efficiency increases with the number of firms in each industry. Lobbying driven markups increase efficiency, but innate markups due to the finite number of firms is purely distortionary.

3.6 Second-Best Analysis

When restricted to designing only the lobbying mechanism parameters, the planner faces a constrained optimization problem:

$$\max_{\{\psi_s, \phi_s, c_s\}} U(C, L) \quad (40)$$

From the perspective of the planner, when constrained to selecting lobbying parameters, two primary trade-offs emerge.

First, the fixed lobbying cost plays a central role. Because lobbying incurs a real fixed cost, the planner ideally sets this cost sufficiently high so that exactly one firm per industry lobbies, especially in industries with multiple firms. However, as individual firm output decreases with the number of firms within an industry, achieving exactly one lobbying firm per industry becomes challenging. Specifically, if an industry with n_j firms supports lobbying by at least one firm, then industries with fewer than n_j firms will likely also contain lobbying firms—often multiple ones. Consequently, achieving exactly one lobbying firm per industry becomes infeasible. This issue can potentially be addressed either by implementing industry-specific lobbying fixed costs or by introducing an artificial auction mechanism allocating subsidies exclusively to a single firm per industry. If these solutions are not available, the planner must carefully balance inter-industry distortions—such as subsidies provided to monopolistic industries—with intra-industry productivity improvements. Targeting one firm to lobby firms around the median industry firm count could represent a practical solution.

Second, planners must weigh the marginal and increasing marginal cost parameters associated with lobbying. Again, the critical balance is between quantity changes driven by the number of firms in an industry versus those driven by productivity differences. Firm quantity influenced by the industry firm count tends to exhibit a high absolute price elasticity (p'_i) when few firms are present, whereas productivity (a_{ij}) does not directly exhibit this dependence. Examining the firm's first-order condition (FOC) with respect to quantity, a fixed lobbying amount l_{ij} increases production quantities more substantially in industries with more firms compared to industries with smaller firms. Thus, planners can strategically exploit this relationship by incentivizing firms whose production quantity increases more significantly in response to lobbying subsidies.

From the firm's lobbying FOC, this approach translates into the following condition:

$$\frac{ds(l_{ij})}{dq_{ij}^*} > 0 \quad (41)$$

This condition implies:

$$\frac{1}{(\phi_s - 1)} s(l_{ij}^*)^{\frac{1}{\phi_s}} \frac{1}{\phi_s \psi_s} > 0 \quad (42)$$

Among the terms above, only $(\phi_s - 1)$ could yield a negative value. This indicates that as ϕ_s approaches but remains greater than one, efficiency gains become more closely tied to firm productivity and output rather than merely the industry's firm count. Consequently, this suggests the planner should predominantly rely on fixed lobbying costs rather than increasing marginal lobbying costs. However, caution is necessary since distortions scale exponentially with lower values of ϕ_s .

4 Computational Method

The model features a finite industry finite firm general equilibrium framework with heterogeneous firms. Thirty-two industries are modeled, each containing a random number of firms drawn from an exponential distribution with a minimum of 1 firm and mean of 6.1 firms, matching the Compustat firm size distribution in 2016. Firm productivities follow an independent Pareto distribution with minimum productivity normalized to 1 and tail parameter ξ , consistent with standard assumptions in the markup literature.

The equilibrium concept combines Nash and general equilibrium. Nash equilibrium requires simultaneous optimization by all firms and the final goods producer. Factor markets clear through endogenous wage determination and a rental rate pinned down by steady-state capital stock conditions. Final output is normalized to one for calibration purposes.

5 Calibration Targets

The model parameters are disciplined through a combination of direct assignment and moment matching, as shown in Tables 1 and 2. Key structural parameters follow standard values from the macroeconomics literature, while industry-specific parameters are calibrated to match empirical moments from firm-level data.

Note: $\bar{\mu}$ target from Delocker et. al. 2020. All values calibrated to match empirical moments. Note: ϕ_l follows Whalen and Reichling (2017) estimates.

Table 1: Assigned Parameter Values

Parameter	Description	Value
β	Discount factor	0.96
δ	Depreciation rate	0.6
ϕ_l	Inverse Frisch elasticity	1/0.44
α	Labor share of income	2/3

Table 2: Calibration Targets and Values

Parameter	Interpretation	Target	Value
ξ	Pareto tail of Productivity	Intra-industry μ_{ij} dist.	1.63
σ	Elasticity of substitution	$\bar{\mu} = 1.67$	1.1
c_s	Fixed lobbying cost	10% lobby part.	1.6×10^{-9}
ψ_s	Marginal lobbying cost	1.7×10^{-4} (lobby/GDP)	2×10^{-5}
ϕ_s	Lobbying marginal returns	Lobby dist.	2
ψ_l	Labor disutility	$Y = 1$	4.646e12

6 Results

The model is still poorly calibrated. Preliminary results indicate a 1.4% increase in aggregate output when lobbying is enacted vs not, suggesting potential net productivity gains. The markup distribution as a result of lobbying increases the markup of most productive firms at the cost of lower markup of the medium productive firms, as shown in Figure 1.

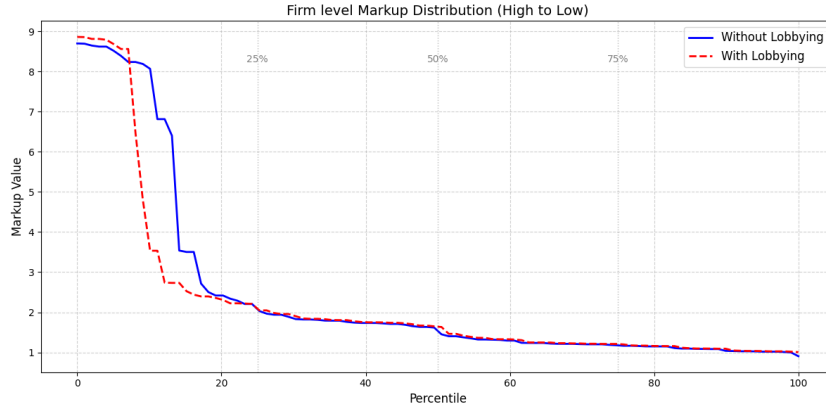


Figure 1: Equilibrium Markup Distribution With and Without Lobbying

Table 3 compares model moments with empirical targets. Calibration requires more time.

Table 3: Calibration Performance

Moment	Target	Model
Mean markup	1.67	1.55
Top 5% intra-industry markup	2.62	2.38
Top 20% intra-industry markup	2.33	2.26
Median intra-industry markup	2.11	1.99
Lobbying firm prevalence	12%	4%
Lobbying/sales ratio	1.68e-4	6.2e-6
Lobbying concentration ratios	Various	1 (degenerate)

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